

Layer 1 • Multi-scale acquisition

Ground / IoT
soil & weather

Proximal
robot-mounted

UAV
multispectral

Satellite
optical / SAR

Layer 2 • Cross-scale alignment
registration • resampling • harmonisation

Layer 3 • Hybrid fusion engine
ML + Kalman/Bayesian • **calibrated uncertainty**

Layer 4 • Dynamic digital twin
assimilation • simulation • forecasting

Layer 5 • Agricultural-robot decisions
field validation • real-world deployment

*active
sensing
feedback*